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$$\begin{aligned}
& \frac{1}{16\pi^2} \left( \frac{1}{108} \frac{1}{m_e^2} s_Y^2 (\overline{y_d^{pi3}} y_d^{pi4} (54 c_Y^2 \overline{y_e^{ri1}} y_e^{ri2} + g_1^2 \delta_{i1i2}) - 5 g_1^2 \overline{y_e^{pi1}} y_e^{pi2} \delta_{i3i4}) + \right. \\
& \frac{4}{81} \sum_p g_1^4 LF_{3,0}[m_d^p] \delta_{i1i2} \delta_{i3i4} - \frac{5}{54} \sum_p g_1^4 LF_{4,-1}[m_d^p] \delta_{i1i2} \delta_{i3i4} + \\
& \frac{16}{405} \sum_p g_1^4 LF_{5,-2}[m_d^p] \delta_{i1i2} \delta_{i3i4} + \frac{4}{27} \sum_p g_1^4 LF_{3,0}[m_e^p] \delta_{i1i2} \delta_{i3i4} - \\
& \frac{5}{18} \sum_p g_1^4 LF_{4,-1}[m_e^p] \delta_{i1i2} \delta_{i3i4} + \frac{16}{135} \sum_p g_1^4 LF_{5,-2}[m_e^p] \delta_{i1i2} \delta_{i3i4} + \\
& \frac{2}{27} \sum_p g_1^4 LF_{3,0}[m_l^p] \delta_{i1i2} \delta_{i3i4} - \frac{5}{36} \sum_p g_1^4 LF_{4,-1}[m_l^p] \delta_{i1i2} \delta_{i3i4} + \\
& \frac{8}{135} \sum_p g_1^4 LF_{5,-2}[m_l^p] \delta_{i1i2} \delta_{i3i4} + \frac{2}{81} \sum_p g_1^4 LF_{3,0}[m_q^p] \delta_{i1i2} \delta_{i3i4} - \\
& \frac{5}{108} \sum_p g_1^4 LF_{4,-1}[m_q^p] \delta_{i1i2} \delta_{i3i4} + \frac{8}{405} \sum_p g_1^4 LF_{5,-2}[m_q^p] \delta_{i1i2} \delta_{i3i4} + \frac{16}{81} \sum_p g_1^4 LF_{3,0}[m_u^p] \\
& \delta_{i1i2} \delta_{i3i4} - \frac{10}{27} \sum_p g_1^4 LF_{4,-1}[m_u^p] \delta_{i1i2} \delta_{i3i4} + \frac{64}{405} \sum_p g_1^4 LF_{5,-2}[m_u^p] \delta_{i1i2} \delta_{i3i4} + \\
& \frac{1}{9} s_Y^2 (\overline{y_d^{pi3}} y_d^{pi4} (9 c_Y^2 \overline{y_e^{ri1}} y_e^{ri2} + g_1^2 \delta_{i1i2}) - g_1^2 \overline{y_e^{pi1}} y_e^{pi2} \delta_{i3i4}) LF_{1,2}[m_\Phi] + \\
& \frac{1}{18} s_Y^2 (3 \overline{y_d^{pi3}} y_d^{pi4} (3 s_Y^2 \overline{y_e^{ri1}} y_e^{ri2} - g_1^2 \delta_{i1i2}) - g_1^2 \overline{y_e^{pi1}} y_e^{pi2} \delta_{i3i4}) LF_{2,1}[m_\Phi] + \\
& \frac{1}{54} g_1^2 (9 s_Y^2 \overline{y_d^{pi3}} y_d^{pi4} \delta_{i1i2} + (3 s_Y^2 \overline{y_e^{pi1}} y_e^{pi2} + 4 g_1^2 \delta_{i1i2}) \delta_{i3i4}) LF_{3,0}[m_\Phi] - \\
& \frac{5}{36} g_1^4 LF_{4,-1}[m_\Phi] \delta_{i1i2} \delta_{i3i4} + \frac{8}{135} g_1^4 LF_{5,-2}[m_\Phi] \delta_{i1i2} \delta_{i3i4} + \frac{2}{27} g_1^4 LF_{3,0}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \\
& \frac{1}{9} g_1^4 LF_{4,-1}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} - \frac{16}{135} g_1^4 LF_{5,-2}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{162} g_1^4 LF_{2,1,0}[m_1, m_d^{-i3}] \delta_{i1i2} \delta_{i3i4} + \\
& \frac{1}{162} g_1^4 LF_{2,2,-1}[m_1, m_d^{-i3}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{81} g_1^4 LF_{3,1,-1}[m_1, m_d^{-i3}] \delta_{i1i2} \delta_{i3i4} + \\
& \frac{1}{162} g_1^4 LF_{4,1,-2}[m_1, m_d^{-i3}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{162} g_1^4 LF_{2,1,0}[m_1, m_d^{-i4}] \delta_{i1i2} \delta_{i3i4} + \\
& \frac{1}{162} g_1^4 LF_{2,2,-1}[m_1, m_d^{-i4}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{81} g_1^4 LF_{3,1,-1}[m_1, m_d^{-i4}] \delta_{i1i2} \delta_{i3i4} + \\
& \frac{1}{162} g_1^4 LF_{4,1,-2}[m_1, m_d^{-i4}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{18} g_1^4 LF_{2,1,0}[m_1, m_e^{-i1}] \delta_{i1i2} \delta_{i3i4} + \\
& \frac{1}{18} g_1^4 LF_{2,2,-1}[m_1, m_e^{-i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{9} g_1^4 LF_{3,1,-1}[m_1, m_e^{-i1}] \delta_{i1i2} \delta_{i3i4} + \\
& \frac{1}{18} g_1^4 LF_{4,1,-2}[m_1, m_e^{-i1}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{18} g_1^4 LF_{2,1,0}[m_1, m_e^{-i2}] \delta_{i1i2} \delta_{i3i4} + \\
& \frac{1}{18} g_1^4 LF_{2,2,-1}[m_1, m_e^{-i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{9} g_1^4 LF_{3,1,-1}[m_1, m_e^{-i2}] \delta_{i1i2} \delta_{i3i4} + \\
& \frac{1}{18} g_1^4 LF_{4,1,-2}[m_1, m_e^{-i2}] \delta_{i1i2} \delta_{i3i4} + \frac{2}{27} g_1^2 g_3^2 LF_{2,1,0}[m_3, m_d^{-i3}] \delta_{i1i2} \delta_{i3i4} + \\
& \frac{2}{27} g_1^2 g_3^2 LF_{2,2,-1}[m_3, m_d^{-i3}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{27} g_1^2 g_3^2 LF_{3,1,-1}[m_3, m_d^{-i3}] \delta_{i1i2} \delta_{i3i4} + \\
& \frac{2}{27} g_1^2 g_3^2 LF_{4,1,-2}[m_3, m_d^{-i3}] \delta_{i1i2} \delta_{i3i4} + \frac{2}{27} g_1^2 g_3^2 LF_{2,1,0}[m_3, m_d^{-i4}] \delta_{i1i2} \delta_{i3i4} + \\
& \frac{2}{27} g_1^2 g_3^2 LF_{2,2,-1}[m_3, m_d^{-i4}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{27} g_1^2 g_3^2 LF_{3,1,-1}[m_3, m_d^{-i4}] \delta_{i1i2} \delta_{i3i4} + \\
& \frac{2}{27} g_1^2 g_3^2 LF_{4,1,-2}[m_3, m_d^{-i4}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{81} g_1^4 LF_{2,1,0}[m_d^{-i3}, m_1] \delta_{i1i2} \delta_{i3i4} + \\
& \frac{1}{162} g_1^4 LF_{3,1,-1}[m_d^{-i3}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{4}{27} g_1^2 g_3^2 LF_{2,1,0}[m_d^{-i3}, m_3] \delta_{i1i2} \delta_{i3i4} + \\
& \frac{2}{27} g_1^2 g_3^2 LF_{3,1,-1}[m_d^{-i3}, m_3] \delta_{i1i2} \delta_{i3i4} - \frac{1}{81} g_1^4 LF_{2,1,0}[m_d^{-i4}, m_1] \delta_{i1i2} \delta_{i3i4} + \\
& \frac{1}{162} g_1^4 LF_{3,1,-1}[m_d^{-i4}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{4}{27} g_1^2 g_3^2 LF_{2,1,0}[m_d^{-i4}, m_3] \delta_{i1i2} \delta_{i3i4} + \\
& \frac{2}{27} g_1^2 g_3^2 LF_{3,1,-1}[m_d^{-i4}, m_3] \delta_{i1i2} \delta_{i3i4} - \frac{1}{9} g_1^4 LF_{2,1,0}[m_e^{-i1}, m_1] \delta_{i1i2} \delta_{i3i4} + \\
& \frac{1}{18} g_1^4 LF_{3,1,-1}[m_e^{-i1}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{1}{9} g_1^4 LF_{2,1,0}[m_e^{-i2}, m_1] \delta_{i1i2} \delta_{i3i4} + \\
& \frac{1}{18} g_1^4 LF_{3,1,-1}[m_e^{-i2}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{1}{9} g_1^2 \overline{y_e^{pi1}} y_e^{pi2} LF_{2,1,0}[m_l^p, \tilde{\mu}] \delta_{i3i4} + \\
& \frac{1}{18} g_1^2 \overline{y_e^{pi1}} y_e^{pi2} LF_{2,2,-1}[m_l^p, \tilde{\mu}] \delta_{i3i4} + \frac{1}{18} g_1^2 \overline{y_e^{pi1}} y_e^{pi2} LF_{3,1,-1}[m_l^p, \tilde{\mu}] \delta_{i3i4} + \\
& \frac{1}{9} g_1^2 \overline{y_d^{pi3}} y_d^{pi4} LF_{2,1,0}[m_q^p, \tilde{\mu}] \delta_{i1i2} - \frac{1}{18} g_1^2 \overline{y_d^{pi3}} y_d^{pi4} LF_{2,2,-1}[m_q^p, \tilde{\mu}] \delta_{i1i2} - \\
& \frac{1}{18} g_1^2 \overline{y_d^{pi3}} y_d^{pi4} LF_{3,1,-1}[m_q^p, \tilde{\mu}] \delta_{i1i2} + \frac{1}{18} g_1^2 \overline{y_e^{pi1}} y_e^{pi2} LF_{2,1,0}[\tilde{\mu}, m_l^p] \delta_{i3i4} - \\
& \frac{5}{18} g_1^2 \overline{y_e^{pi1}} y_e^{pi2} LF_{3,1,-1}[\tilde{\mu}, m_l^p] \delta_{i3i4} + \frac{1}{9} g_1^2 \overline{y_e^{pi1}} y_e^{pi2} LF_{4,1,-2}[\tilde{\mu}, m_l^p] \delta_{i3i4} - \\
& \frac{1}{18} g_1^2 \overline{y_d^{pi3}} y_d^{pi4} LF_{2,1,0}[\tilde{\mu}, m_q^p] \delta_{i1i2} - \frac{7}{18} g_1^2 \overline{y_d^{pi3}} y_d^{pi4} LF_{3,1,-1}[\tilde{\mu}, m_q^p] \delta_{i1i2} + \\
& \frac{1}{9} g_1^2 \overline{y_d^{pi3}} y_d^{pi4} LF_{4,1,-2}[\tilde{\mu}, m_q^p] \delta_{i1i2} + \frac{1}{18} g_1^4 LF_{2,1,1,-1}[m_1, m_d^{-i3}, m_e^{-i1}] \delta_{i1i2} \delta_{i3i4} + \\
& \frac{1}{9} g_1^4 m_1^2 LF_{2,1,1,0}[m_1, m_d^{-i3}, m_e^{-i1}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{18} g_1^4 LF_{2,1,1,-1}[m_1, m_d^{-i4}, m_e^{-i2}] \delta_{i1i2} \delta_{i3i4} + \\
& \frac{1}{9} g_1^4 m_1^2 LF_{2,1,1,0}[m_1, m_d^{-i4}, m_e^{-i2}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{2} \overline{y_d^{pi3}} y_d^{pi4} \overline{y_e^{ri1}} y_e^{ri2} LF_{2,1,1,-1}[\tilde{\mu}, m_l^r, m_q^p] \Big)
\end{aligned}$$